

Networks (NW)

Foundations of Networks

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- **Required components of a (computer) network**
 - At least 2 terminal devices
 - Communication intention
 - Network service for communication or shared resource usage
 - Transmission media
 - Wire-based: electrical wires or fibre-optical cables
 - Wireless
 - Network protocol(s)
 - Rules and specification for managing communication

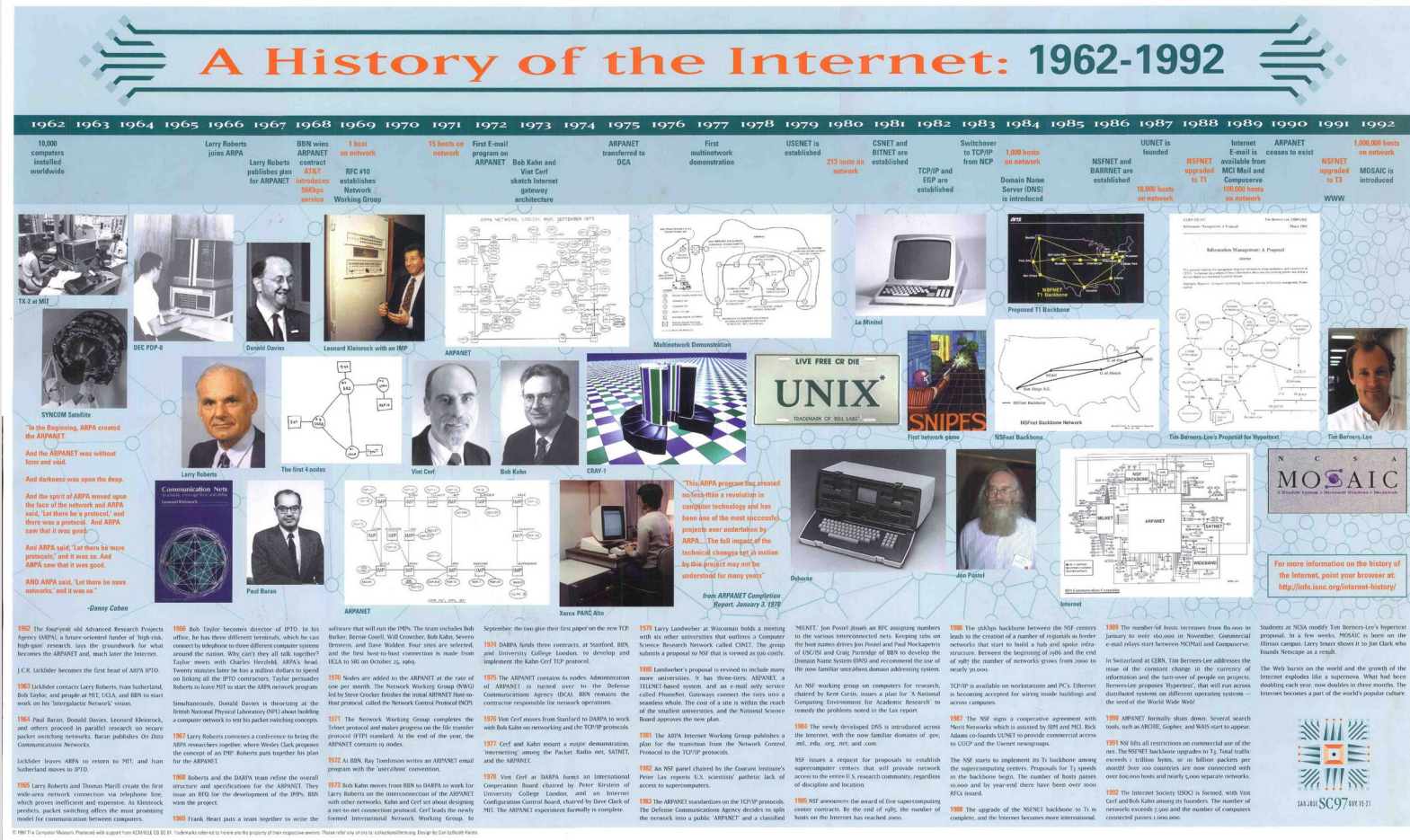
Advanced Research Projects Agency Network (ARPAnet)

- First decentralized computer network
- Connected 4 U.S. Universities in 1969
- Renamed to DARPAAnet in 1971
- Based on (unreliable) phone lines
 - Packet switching
- **In 1970s**
 - Connection of some terminals to mainframe computers (multi-user systems)
- **1974: IBM's System Network Architecture (SNA)**
 -

History of Computer Networks



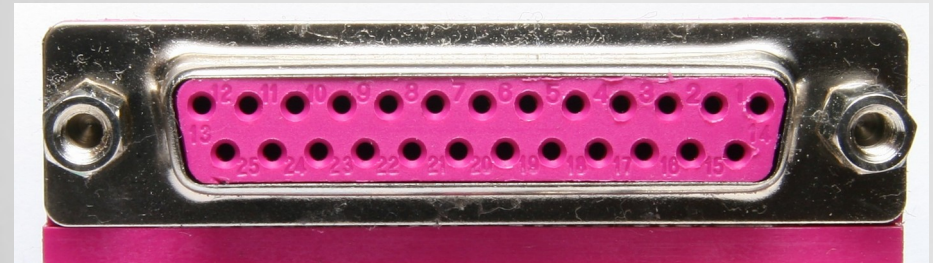
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<https://tcm.computerhistory.org/marketing/posterhistoryofinternet1962-1992.jpg>

Parallel Data Transmission

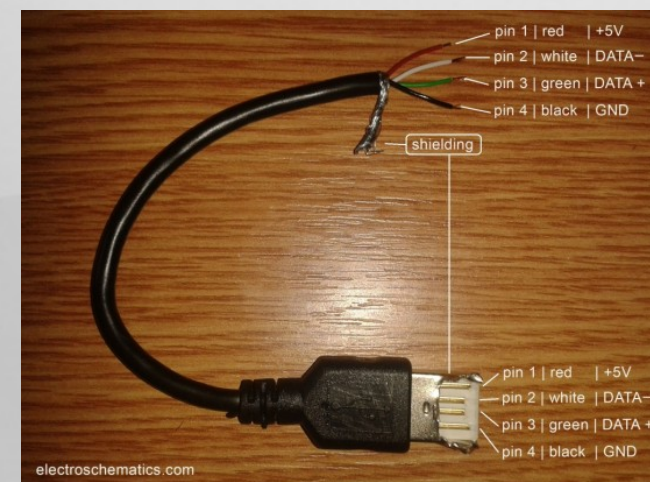
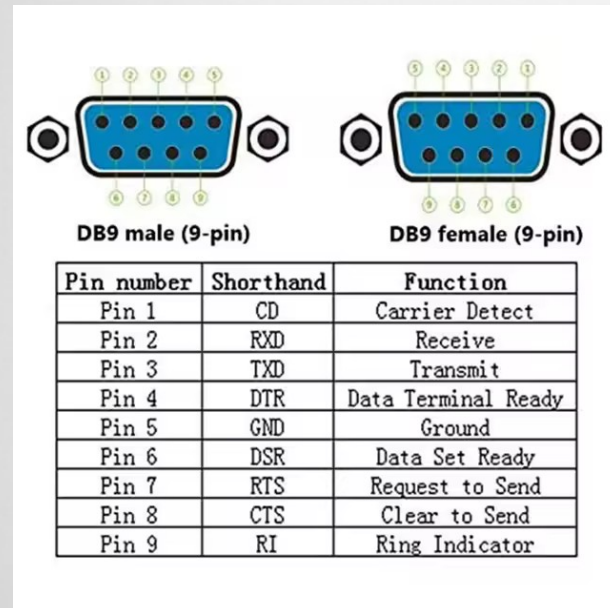
- **Parallel electrical lines**
 - 8 (8 bit in parallel), 16, 32, or 64 plus control lines (Controls, RW, clock)
- Local bus systems
- Higher throughput (bandwidth) compared to serial line
- e.g. Parallel port for printer
- Drawback: Expensive
- **Examples:**
 - ISA, ATA, SCSI, IEEE 1284 (printer port)



<https://upload.wikimedia.org/wikipedia/commons/2/24/Parallelport.jpg>

Serial Data Transmission

- **Single electrical lines**
 - Bit by bit transmission
- Local bus systems and computer networks
- Transmission on longer distances
- Drawback: Less throughput
- **Examples:**
RS 232, 488, USB, CAN, Fibre Channel (for SAN), Ethernet



What are the Pros and Cons of Parallel vs. Serial Data Transmission?



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- **Pros**

- **Parallel**

- **Serial**

- **Cons**

- **Parallel**

- **Serial**

Categories of Network Extensions

- **Personal Area Network (PAN)**
 - few meters
 - Bluetooth, IrDA, Wifi, USB, FireWire
- **Local Area Network (LAN)**
 - Max. 500 – 1000m depending on transmission medium
 - WLAN – IEEE 802.11
 - Ethernet – IEEE 802.3
 - Token Ring – IEEE 802.5
- **Metropolitan Area Network (MAN)**
 - Up to 100km
 - Preferable fibre-optical media
 - WiMAX – IEEE 802.16
- **Wide Area Network (WAN)**
 - Up to 1000km
 - **Asynchronous Transfer Mode (ATM)**
 - 10G Ethernet
- **Global Area Network (GAN)**
 - Nearly unlimited
 - Fibre-optical
 - Satellite based communication

Directional Dependence of Data Transmission

Full-duplex

- The information transfer works in both directions simultaneously
- Examples: Phone, networks with twisted pair cables because they provide
- separate wires for send and receive

Half-duplex

- The information transfer works in both directions
- Only one direction at a time
- Examples: Networks with fiber-optic cables or coaxial cables (a single line to sending and receiving)
- Wireless networks with just a single channel

Simplex

- The information transfer only works in one direction
- After the end of a transmission, the communication channel can be used by another sender

Network Topologies

- **Network topology describe the connection among terminals**
- **Sometimes hybrid topologies, a combination of base topologies**
- **Distinction in**
 - Physical topology
 - Connection type of physical media among terminals
 - Logical topology
 - Control + data flow between terminal
- **Graphical representation by nodes and edges**

Peer-to-peer Connection



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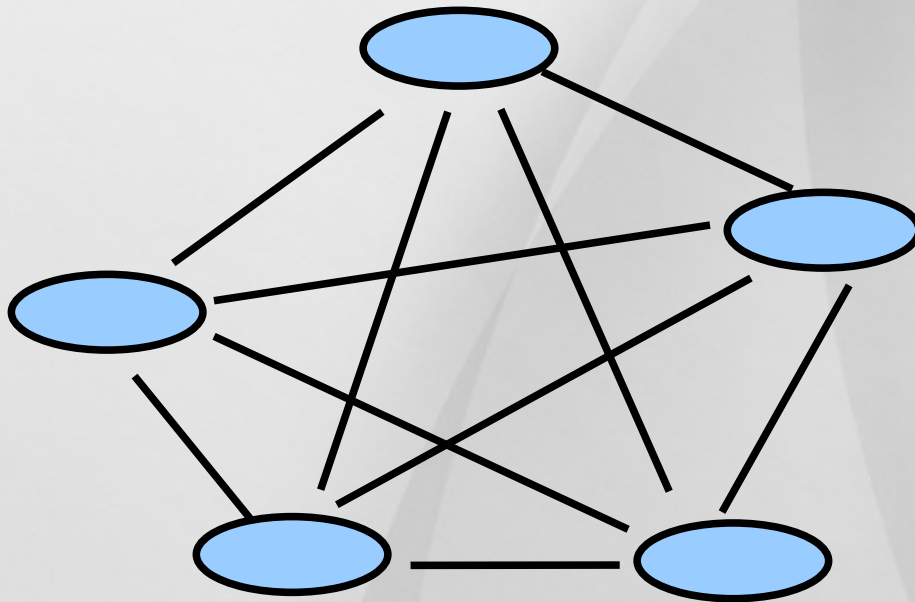
- Direct connection between two nodes
- Uni- and bi-directions
- No central resources



Fully Connected - Mesh



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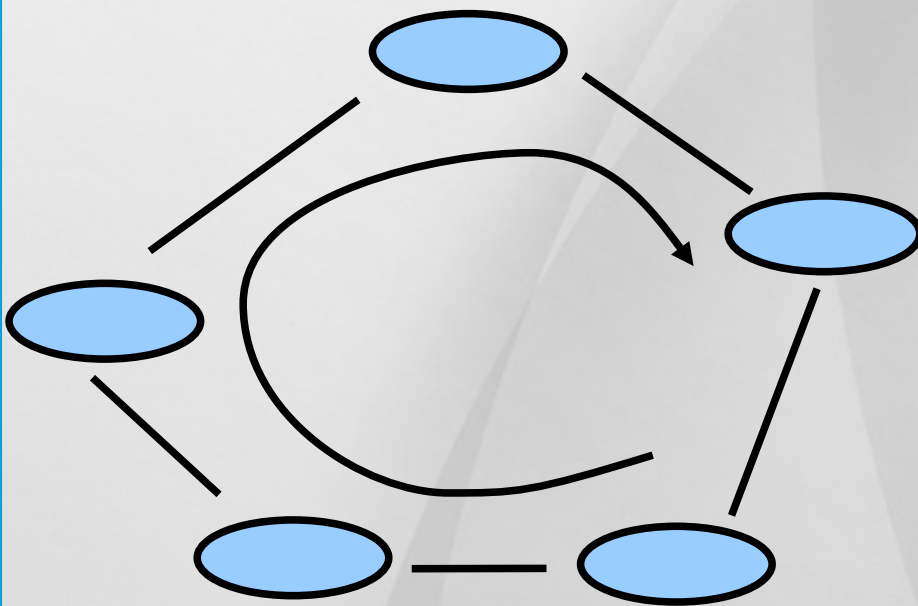


- Extension of peer-to-peer connection
- Each node is connected to $n-1$ other nodes
 - → $n-1$ interfaces per node
 - → $\binom{n}{2} = \frac{n \cdot (n-1)}{2}$ connections
- → many connections (i.e. cables & interfaces) needed
- Redundancy through many paths from node to another using further nodes
- **No practical topology**

Ring



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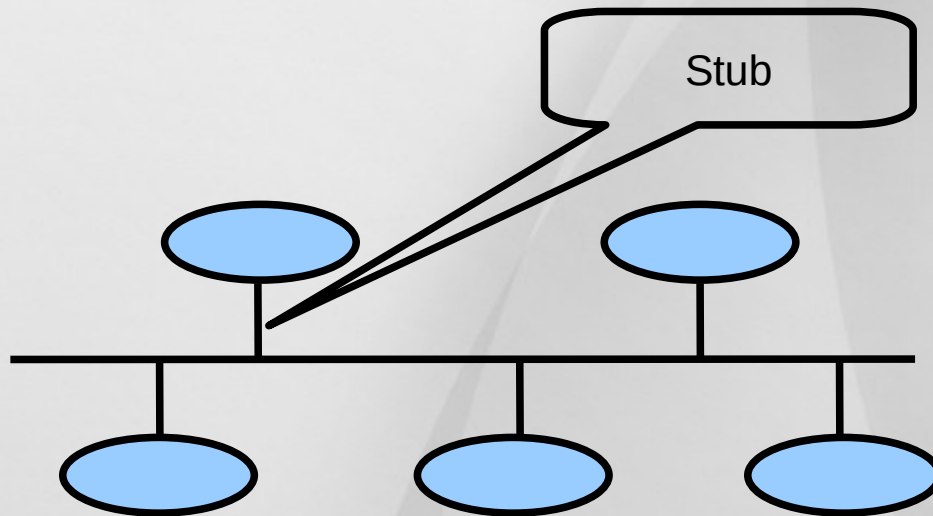


- Point-to-point in 2 directions
- Logical ring of nodes
- All nodes need to be active (both receiving + transmitting)
- Broken wire healed by two way communication

Bus or Linear Structure

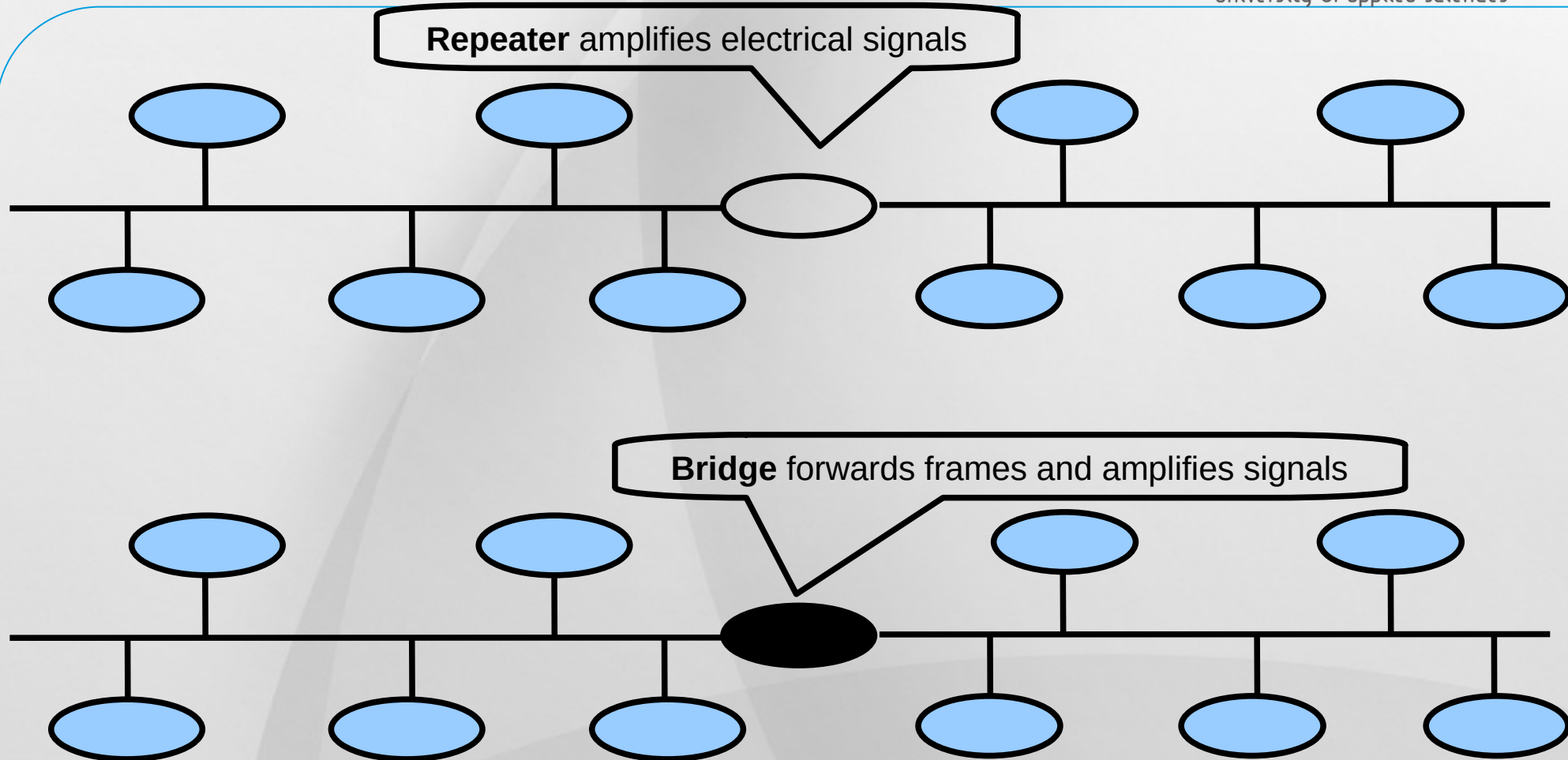


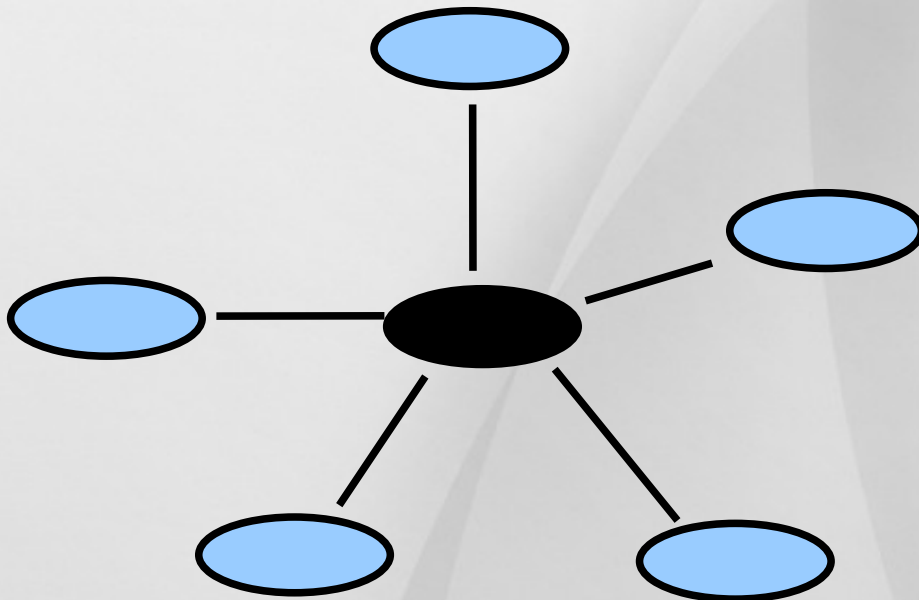
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- Network from one ended node to the other ended node
- All nodes are connected to one or two single lines
- All nodes listen to communication
- A short-cut inhibits all network communication
- A cut divides a bus into two parts, bus segments

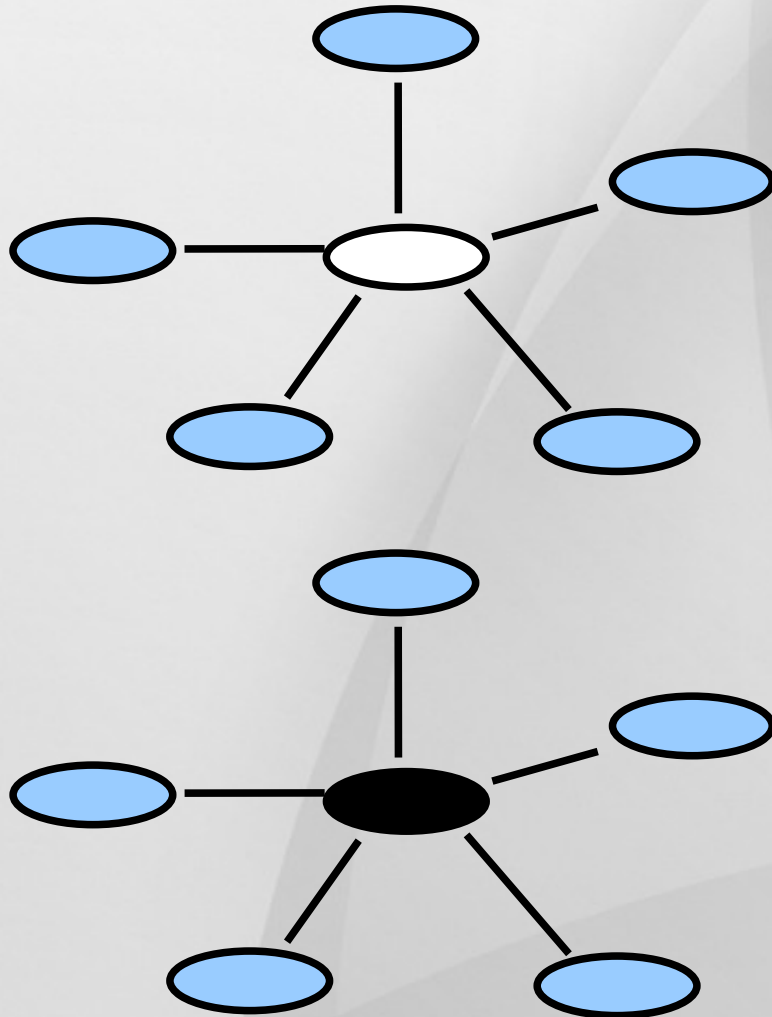
Connecting Busses, Repeater vs. Bridge



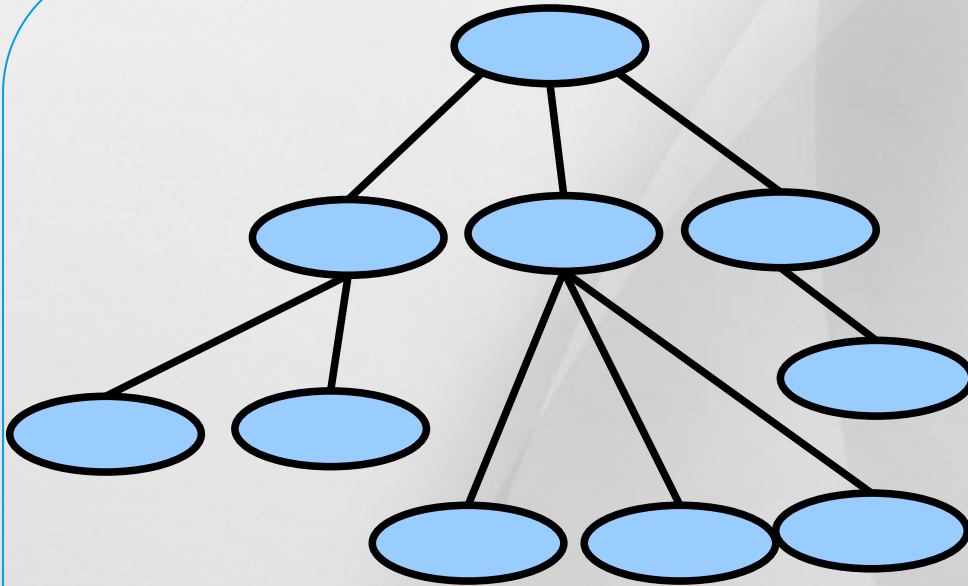


- All nodes connected to a central node (aka. star coupler)
- Central node's working is essential (single point of failure)
- Easy to find failure of single nodes
- Deterministic latency
- Distinction (see Hub/Switch/Router):
 - Active central node
 - Passive central node

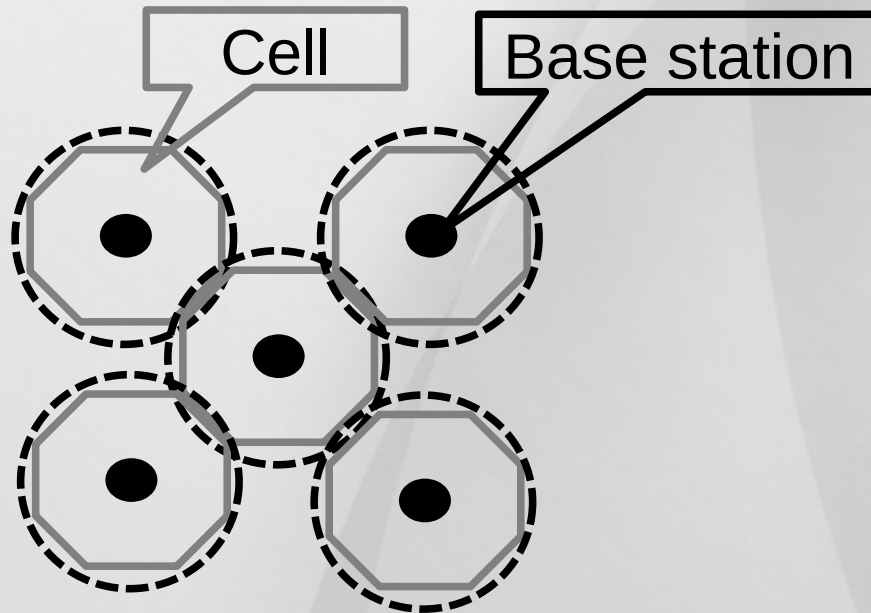
Starcoupler – Hub, Switch, Router



- Hub – Multiport repeater
 - Signal amplification only
- Switch – Multiport Bridge
 - Connects/seperates different physical networks
- Router
 - Logical / Packet based connection



- Hierarchical connections of star topologies
- Long(er) distance transmissions supported
- Leaf failure affects leaf only leaf
- Node failure → sub tree not accessible



- Wireless networks only for establishing a node connection
- All nodes in a cell could connect to the respective base station
- Cell as collision domain – only one sender can send data
- Used by:
 - GSM
 - Wlan – IEEE 802.11
 - Bluetooth

Pros and Cons of the Topologies I

Fully connected - Mesh

Pros

-

-

Cons

-

-

Star

Pros and Cons of the Topologies II

Ring

Pros

-

-

Cons

-

-

Bus

Network Topologies Nowadays

- **Ring**
 - Very rarely used
- **Bus**
 - Industrial & Automotive Area: CAN
 - PowerLAN (Powerline Communication)
- **Star**
 - Ethernet from switch to leaf
- **Tree**
 - USB
 - Among Switches and Hubs using Ethernet
- **Mesh**
 - Special wireless network modes
 - Logical topology connecting routers
 - Rarely used inside computer systems connecting computing nodes (cores)